Listing 5.2 Naïve solution, add

```
function
add(elem)
      #1
  throw-if elem ==
null                                                          #2
  if partitionsMap.has(elem) then
                                          #3
    return false
  partitionsMap[elem] = new
Set(elem)                                       #4
  return true
```

#1 Takes an element and returns `true` iff[8] the element was added successfully, `false` otherwise.
#2 Check that the element is valid.
#3 If the element is already in the data structure, return `false` without updating anything.
#4 Otherwise just add a mapping between the element and a newly created singleton[9] set, and return `true`.

First, we initialize the associative array that is going to index the elements and map them to the partition they belong to (line #1).

Next, we just go through `initialSet`'s elements, one by one, check they are defined and that there is no duplicate, then initialize their partition to the singleton containing the element itself (initially each element is disjoint from every other element).